

Phase 6- Project Testing Phase

Project title : Script-Controlled ACL - Restrict Record Access Based on Field Value

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The Project Testing Phase is the stage where the developed Script-Controlled ACL – Restrict Record Access Based on Field Value system is carefully checked to ensure that all access-control rules work correctly. The main objective is to verify that authorized users can access permitted records and unauthorized users are prevented from accessing restricted records.

Testing is performed using different users, records, roles, and field values. The results are compared with the expected access decisions.

6.1 Functional Testing

Functional testing verifies whether the main functions of the ACL system work according to the requirements. The system is checked for user authentication, record access, field-value checking, and access decisions.

6.2 User Authentication Testing

The login process is tested with valid and invalid user details. The purpose is to ensure that only authenticated users can request protected records.

6.3 Field Value Testing

Different values are entered into the selected control field. The ACL script is checked to ensure that it correctly reads and evaluates each value.

6.4 Authorized User Testing

An authorized user attempts to access a record that satisfies the required condition.

Expected Result: Access should be granted.

6.5 Unauthorized User Testing

A user without the required permission attempts to access a restricted record.

Expected Result: Access should be denied.

6.6 Matching Field Value Testing

The field value is set to a value that satisfies the ACL condition.

Example:

Matching field value → Access Granted

This confirms that the system correctly identifies valid access conditions.

6.7 Non-Matching Field Value Testing

The field value is changed to a value that does not satisfy the ACL condition.

Example:

Non-matching field value → Access Denied

6.8 Role-Based Testing

Different user roles are tested to verify that the ACL applies the correct restrictions to each type of user.

6.9 Record-Level Testing

Different records are tested individually. This ensures that access can be controlled according to the field value of each record rather than applying the same decision to every record.

6.10 Negative Testing

Negative testing checks how the system behaves when incorrect or unauthorized requests are made. Invalid users, incorrect field values, and restricted records are tested.

6.11 Error Handling Testing

The system is tested with empty, missing, invalid, or unexpected field values. The purpose is to ensure that such conditions do not accidentally provide unauthorized access.

6.12 Security Testing

Security testing verifies that users cannot easily bypass the ACL restrictions. The system is checked to ensure that the defined access conditions are consistently enforced.

6.13 Performance Testing

The system is checked to ensure that the ACL validation does not cause unnecessary delays when users request records.

6.14 Regression Testing

After correcting an error in the ACL script, previously tested functions are tested again. This ensures that a new change has not affected existing access-control functionality.

6.15 Bug Identification

Any unexpected result found during testing is recorded as a defect. The relevant ACL script or configuration is then reviewed and corrected.

6.16 Final Validation

After completing all test cases, the actual results are compared with the expected results. If the results match the requirements, the system is considered ready for final implementation and demonstration.

Sample Test Cases

Test Case	Condition	Expected Result
TC01	Valid user + matching field value	Access Granted
TC02	Unauthorized user + matching field value	Access Denied
TC03	Valid user + non-matching field value	Access Denied
TC04	Missing field value	Access Denied
TC05	Invalid user	Access Denied
TC06	Valid user+permitted record	Access Granted

Testing Flow

User Request → Authentication → Identify Record → Check Field Value → Execute ACL → Validate Condition → Compare Result → Grant / Deny Access

Expected Outcome

The testing phase ensures that the Script-Controlled ACL works accurately and consistently. It confirms that record access is properly controlled according to field values and that unauthorized access is prevented.